

Improved Disturbance Rejection for High Precision Systems through Estimation of the Flexible Modes

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Abstract—This paper demonstrates applications of a state observer in high precision control systems that improve low frequency disturbance rejection. By changing the plant transfer function as seen by the feedback controller the effect of flexible modes on the feedback loop is reduced. This allows a redesign of the feedback controller for improved control bandwidth and, consequently, improved low frequency disturbance rejection. Simulation and experimental validation results are presented for the applications to elucidate their (combined) effects.

I. INTRODUCTION

In high precision applications, increasingly stringent performance requirements on acceleration and positioning accuracy no longer allow mechanical systems to be considered as infinitely stiff [1]. An example of such a high precision control system is the wafer stage. This machine is used in photolithography applications to position a silicon wafer underneath a projection lens. The positioning has to be within nanometer accuracy with accelerations up to 5 g to meet the desired throughput requirements [2].

The control forces and disturbances deform the system, which affects the positioning accuracy. The flexible behavior (i.e. vibration) results in anti-resonance and resonance pairs in the system transfer function of the system. The employed sensor system results in anti resonance frequencies that vary with position, limiting the achievable control bandwidth. If the effects of the flexible dynamics are reduced, an increase in control bandwidth and, consequently, improved low frequency disturbance rejection is possible.

To control the flexible behavior accurate state information is required. In [3] a modal filter reconstructs the states of a flexible system. However, modal filters, and other applications of oversensing, require a large number of sensors to produce online measurements of the deformed structure. Because high precision systems typically have an insufficient number of sensors to apply those methods, an observer based approach is used in this paper.

In general, an observer based approach utilizes full state feedback [4], [5]. Such an approach works well for vibration reduction in, for instance, lightweight space structures, but it is unsuited for high precision applications. The estimated rigid body mode cannot be used for feedback as the observer introduces phase delay. This leads to performance deterioration as a delayed version of the true state is controlled. Furthermore, the effects of input and output noise result in an estimate that is worse than the position measurement

provided by the sensors. Therefore, we propose to add an observer to the existing position controller of Fig. 1, to reduce the effect of flexible modes.

Two applications of the observer are proposed. The first is compensating the measurements for the effect of the flexible dynamics. This approach resembles the disturbance observer [6], where an observer estimates disturbances affecting the plant input such that they can be compensated. Here, the effect of flexible dynamics contaminating the plant output are estimated and used to correct the measurements. The second approach is linked to modal space control [7]. By closing a partial state feedback loop around the plant, the modal damping of certain flexible modes is increased.

To keep in line with high precision flexible systems that will benefit from our results, a single feedback controller is used that stabilizes the flexible system for all possible fixed positions. The position dependent observability of the modes is disregarded to focus on the effect of the observer. The results are easily extendable to the position dependent system by employing a position dependent observer, see for instance [8] for a state dependent Kalman filter.

The final result, which consists of a feedback controller and an observer, can be seen as a new two degrees of freedom LTI controller when the position dependent effects are disregarded. Such a, higher order controller could also be obtained by optimization based methods like H_∞ [9]. However, our approach uses the unfiltered position sensor signal and the tuning parameters are explicitly based on physical properties like noise levels, mode selection and damping. In H_∞ the tuning parameters are weighting filters which are implicitly related to physical properties. Furthermore, extension to position dependent dynamics is straightforward in our method, whereas this is more involved in H_∞ [10].

II. PROBLEM FORMULATION

A typical control loop for high precision positioning systems consists of a feedback controller, $\Sigma_{C_{fb}}$, and a feedforward controller, $\Sigma_{C_{ff}}$, connected to the plant, Σ_P , as seen in Fig. 1. The feedback controller is typically a

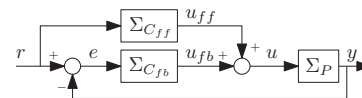


Fig. 1: Typical control configuration

PID controller with additional filters [2] that stabilizes the system and provides low frequency disturbance rejection. The feedforward controller accounts for the majority of the reference tracking performance [11]. The feedforward

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controller is disregarded in this paper as the focus is on disturbance rejection. When the results in this paper are applied to a system where reference tracking is important, the existing feedforward controller must be redesigned to account for the changed transfer from control effort to tracking error.

The goal is increasing the low frequency disturbance rejection. This is achieved by applying an observer in the loop such that the feedback control bandwidth can be increased. A Kalman observer is used for simulation and experimental validation. The state estimates are used to modify the transfer function seen by the feedback controller, resulting in mitigation of the unwanted flexible effects.

III. SYSTEM DESCRIPTION

A state space model in modal form facilitates the state estimation for flexible systems. This form is obtained by performing an eigen decomposition on the equations of motion [7]. In modal form (1) the modes of the system are described by individually decoupled second order systems. As such, the state vector consists of the modal position η_i and modal rate of change $\dot{\eta}_i$ for each mode $i \in \{1, \dots, N\}$.

$$\begin{bmatrix} \dot{\eta}_1 \\ \ddot{\eta}_1 \\ \vdots \\ \dot{\eta}_N \\ \ddot{\eta}_N \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 0 & 1 \\ -\omega_1 & -\zeta_1 \end{bmatrix} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \begin{bmatrix} 0 & 1 \\ -\omega_N & -\zeta_N \end{bmatrix} \end{bmatrix} \begin{bmatrix} \eta_1 \\ \dot{\eta}_1 \\ \vdots \\ \eta_N \\ \dot{\eta}_N \end{bmatrix} + \begin{bmatrix} 0 \\ \phi_{a1}^\top \\ \vdots \\ 0 \\ \phi_{aN}^\top \end{bmatrix} u \quad (1)$$

where $\mathbf{0}$ denotes a zero matrix of appropriate dimensions. Here ω_i and ζ_i denote the natural frequency and modal damping of mode i respectively. $\phi_{ai}^\top$ denotes the coupling between input u and mode i and is determined by evaluating the spatial mode shape ϕ_i at the actuator position. There exists a natural distinction between the desired rigid body modes and the undesired flexible modes. For the rigid body modes both ω_i and ζ_i are zero. In this paper the ordering $0 = \omega_1 \leq \omega_2 \leq \dots \leq \omega_N$ is used.

The output of the system is the combined effect of all modes evaluated at the sensor location. The gain between mode i and the sensors is determined by evaluating the mode shape ϕ_i at the sensor location, hereafter denoted as ϕ_{si} . The measurements y are then given by,

$$y = \sum_{i=1}^N \phi_{si} \eta_i = [\phi_{s1} \ 0 \ \dots \ \phi_{sN} \ 0] \eta \quad (2)$$

where $\eta = [\eta_1 \ \dot{\eta}_1 \ \dots \ \eta_N \ \dot{\eta}_N]^\top \in \mathbb{R}^{2N}$. Combining (1) and (2) results in the state space model in modal form,

$$\Sigma_P \begin{cases} \dot{\eta} &= A\eta + Bu + Gw \\ y &= C\eta + v \end{cases} \quad (3)$$

Where $A \in \mathbb{R}^{2N \times 2N}$, $B \in \mathbb{R}^{2N \times p}$, $C \in \mathbb{R}^{m \times 2N}$. The matrix $G \in \mathbb{R}^{2N \times \dim(w)}$ denotes the effect of the Gaussian, white, zero mean disturbances $w(t)$ on the state dynamics. Without loss of generality, additive noise on the control input u is assumed, thus $G = B$ and $\dim(w) = p$. The measurement noise $v(t) \in \mathbb{R}^m$ consists of m independent Gaussian, white, zero mean noise signals that affect the sensors.

TABLE I: Numerical values used in the flexible plant model

Description	Symbol	Value
Actuator mass	m_a	2.5 [kg]
Mass of a single element	m	$\frac{10}{7}$ [kg]
Actuator connection stiffness	k_a	1e8 [N/m]
Internal stiffness	k	1.12e8 [N/m]
Damping	b, b_a	Tuned for 0.5% modal damping

A. Example system

The flexible system of Fig. 2 is used to show the applications and perform the simulations of Section VI. The system contains eight modes, one rigid body mode and seven flexible modes. Output y_5 is used to measure the position of the system. The numerical values are given in Table I.

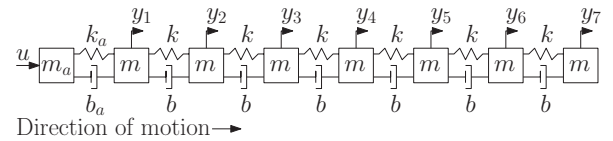


Fig. 2: The SISO flexible system used for simulation.

The feedback controller follows the structure used in [2], where a PID plus low pass filter is employed. In addition, a notch filter is added to cancel one resonant mode. The transfer function of the controller is given by,

$$C_{fb}(s) = K_p \frac{s + \omega_i}{s} \left(\frac{s}{\omega_d} + 1 \right) \frac{\omega_{lp}^2}{s^2 + 2z_{lp}\omega_{lp}s + \omega_{lp}^2} \cdot \frac{s^2 + 2z_r\omega_r s + \omega_r^2}{s^2 + 2z_d\omega_r s + \omega_r^2} \quad (4)$$

and the parameters depend on the desired bandwidth $\omega_{bw} = 2\pi f_{bw}$, according to Table II. The bandwidth is maximized under the constraint that the resonance peaks in PC_{fb} remain below -6 dB for all measurement locations, y_i , $i = 1, \dots, 7$, in Fig. 2. The notch frequency is chosen equal to the dominant resonant mode. Here, dominant resonant mode refers to the flexible mode, whose resonance is the first to violate the -6 dB bound on PC_{fb} . This procedure yields a bandwidth (0dB crossing of the open loop) of at least 39 Hz for all measurement locations.

IV. OBSERVER DESIGN CONSIDERATIONS

The flexible system (3) has $2N$ states. The first few flexible modes are the limiting factor for control bandwidth. Therefore, the reduced observer model contains only $n < N$ modes, the rigid body mode and $n - 1$ flexible modes. From the structure in (1) it follows that model (3) can easily be truncated. Let,

$$\Sigma_{P_t} \begin{cases} \dot{\eta}_t &= A_t \eta_t + B_t u + G_t w \\ y_t &= C_t \eta_t + v \end{cases} \quad (5)$$

denote the truncated version of (3) where $\eta_t = [\eta_1 \ \dot{\eta}_1 \ \dots \ \eta_n \ \dot{\eta}_n]^\top \in \mathbb{R}^{2n}$ and A_t, B_t, C_t, G_t are truncated versions of the system matrices in (3).

TABLE II: [2] Parameter values for the controller

Symbol	Meaning	Value
ω_{bw}	Desired bandwidth	$2\pi f_{bw}$ [rad/s]
M	Total mass = $7m + m_a$	12.5 [kg]
K_p	Proportional gain	$\frac{M}{3}\omega_{bw}^2$
ω_i	Integrator frequency	$\frac{1}{9}\omega_{bw}$ [rad/s]
ω_d	Differentiator frequency	$\frac{1}{3}\omega_{bw}$ [rad/s]
ω_{lp}	Lowpass filter frequency	$3\omega_{bw}$ [rad/s]
z_{lp}	Lowpass filter damping ratio	0.7
ω_r	Notch frequency	Chosen at dominant resonance frequency
z_r	Notch numerator damping ratio	0.001
z_d	Notch denominator damping ratio	0.1

Any model mismatch between Σ_P and Σ_{P_t} is mapped to the estimated states. Thus, the estimation error will be large at the resonance frequencies of the discarded modes. Therefore, the selection of the n modes is important. No discarded mode should have a natural frequency close to the desired bandwidth.

There is a secondary effect to take into account. The compliance of the discarded modes still affects the system at low frequencies. In particular, a force applied to the system results in a compression, due to the combined compliance effect of all modes. The observer gain aims to bring $e_{\text{obs}} = y - \hat{y}$ to zero. Thus, the compliance effect of the discarded modes is mapped onto the state estimates, which yields erroneous results. Therefore, the observer model is augmented to capture the compliance effect of the discarded modes. The augmentation procedure is straightforward: addition of a direct feedthrough matrix decreases the model error in the frequency region of interest. Denote the discarded dynamics, Σ_{P_d} , with state space matrices (A_d, B_d, C_d) . The DC gain is then given by $T_{\text{DC}} = -C_d A_d^{-1} B_d$. The matrix T_{DC} is added to (5) to obtain the augmented model,

$$\Sigma_{\hat{P}} \begin{cases} \dot{\hat{\eta}}_t &= A_t \hat{\eta}_t + B_t u + G_t w \\ y_t &= C_t \hat{\eta}_t + T_{\text{DC}} u + v \end{cases} \quad (6)$$

The effectiveness of the augmentation is shown in Fig. 3. Here, the flexible system is shown (black). The system contains a single rigid body mode and seven flexible modes. The model error between the real plant (3) and the truncated model (5) that contains the rigid body mode and three flexible modes is depicted as well (red). Finally, the model error between the real plant (3) and augmented model (6) is shown (blue). It is clear that the low frequency model error is greatly reduced. The high frequency model error increases due to the augmentation, but this is not an issue for our application.

A Kalman observer (7) is designed for the augmented model (6). Here, L is the observer gain matrix that follows from the algebraic Riccati equation. The value of L depends directly on the input and output noise covariance matrices which are clear physical parameters.

$$\Sigma_O \begin{cases} \dot{\hat{\eta}}_t &= A_t \hat{\eta}_t + B_t u + L(y - C_t \hat{\eta}_t - T_{\text{DC}} u) \\ &= (A_t - LC_t) \hat{\eta}_t + B_t u + L(y - T_{\text{DC}} u) \end{cases} \quad (7)$$

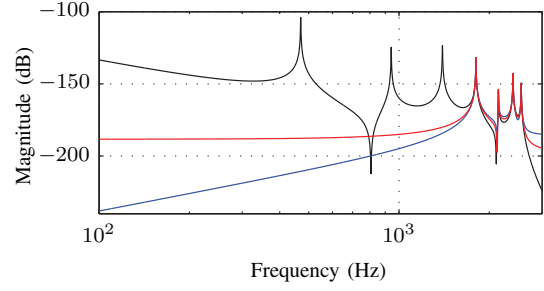


Fig. 3: Real plant (Σ_P) (black) and model error with ($\Sigma_P - \Sigma_{\hat{P}}$) (blue) and without ($\Sigma_P - \Sigma_{P_t}$) (red) a compliance compensation.

Note the usage of y in (7) instead of y_t as the observer takes measurements of the true, full plant (3).

V. COMPENSATION USING OBSERVER IN THE LOOP

In this section two applications of the observer in addition to the existing control loop of Fig. 1, and their effect on the performance are discussed.

The first application is compensating the measurement, $\tilde{y} = y - \hat{y}_f$ for the estimated flexible effects, $\hat{y}_f = C_f \hat{\eta}_t$. Here C_f denotes the matrix C_t in (6) where the column that corresponds to the rigid body mode is zero. The design freedom available is the selection of flexible modes to be removed from the measured output.

The second application is closing an additional loop around the plant to apply a damping force $u_d = K\hat{\eta}$. The matrix $K \in \mathbb{R}^{p \times 2n}$ is a matrix with entries only on the columns corresponding to the rate of change of the modes whose damping is increased. Fig. 4 shows how each application is integrated in the existing control loop.

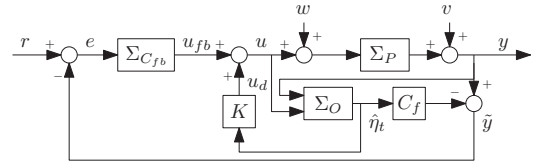


Fig. 4: The control loop with observer applications that reduce flexible effects.

First, it is shown that stability of the closed loop including the observer can be evaluated a priori. Subsequently, the effect of both applications individually is shown. Finally, the joint application of both approaches is discussed. The interconnection of plant, controller and observer of Fig. 4 is displayed in a slightly reordered manner in Fig. 5. The plant dynamics have been split up in a part denoting the retained modes and a part denoting the discarded modes. The discarded modes and the compliance compensation have been extracted and make up the block S with input u_S and output y_S . The observer, truncated plant and feedback controller are grouped in the nominal plant H which is partitioned as,

$$\begin{bmatrix} u_S \\ y \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} y_S \\ r \\ w \\ v \end{bmatrix} \quad (8)$$

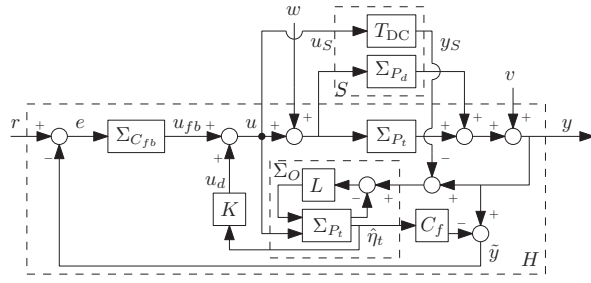


Fig. 5: The control loop with T_{DC} and discarded dynamics Σ_{P_d} modeled as perturbations.

The total transfer from $[r \ w \ v]^T$ to y is given by the upper fractional representation,

$$F_U(H, S) = H_{22} + H_{21}S(I - H_{11}S)^{-1}H_{12} \quad (9)$$

First, consider the case where $S = 0$, i.e. there are no discarded modes and, consequently, T_{DC} is zero. Let the triple (A_c, B_c, C_c) denote the state space matrices of the feedback controller $\Sigma_{C_{fb}}$ with state x_c . The nominal closed loop system, $F_U(H, 0) = H_{22}$, is then given by,

$$\begin{aligned} \begin{bmatrix} \dot{\eta}_t \\ \dot{x}_c \\ \dot{e}_t \end{bmatrix} &= \begin{bmatrix} A_t + B_t K & B_t C_c & -B_t K \\ -B_c(C_t - C_f) & A_c & -B_c C_f \\ \mathbf{0} & \mathbf{0} & A_t - LC_t \end{bmatrix} \begin{bmatrix} \eta_t \\ x_c \\ e_t \end{bmatrix} \\ &+ \begin{bmatrix} \mathbf{0} & G_t & \mathbf{0} \\ B_c & \mathbf{0} & -B_c \\ \mathbf{0} & G_t & -L \end{bmatrix} \begin{bmatrix} r \\ w \\ v \end{bmatrix} \\ y &= [C_t \ \mathbf{0} \ \mathbf{0}] \begin{bmatrix} \eta_t \\ x_c \\ e_t \end{bmatrix} + [\mathbf{0} \ \mathbf{0} \ I] \begin{bmatrix} r \\ w \\ v \end{bmatrix} \end{aligned} \quad (10)$$

where $e_t = \eta_t - \hat{\eta}_t$ denotes the estimation error.

The A matrix in (10) is triangular and thus, the observer and compensation matrices K and C_f can be designed separately. Provided that the diagonal blocks of (10) are individually stable the closed loop will be stable as well.

In practice, discarded dynamics are present and $T_{DC} \neq 0$. In that case the A matrix of $F_U(H, S)$ no longer has a triangular structure. The small gain theorem is used to investigate stability in the presence of these perturbations. When all blocks H_{ij} are individually stable, instability of the full system (9) can only occur when $\|H_{11}S\| \geq 1$. Once K , C_f and the observer are designed the condition $\|H_{11}S\|_\infty < 1$ is checked to ensure stability. For all applications considered in this paper, the closed loop is stable in the presence of the discarded modes and the compliance compensation T_{DC} .

A. Measurement correction

By subtracting the estimated flexible effect from the measurements, $\tilde{y} = y - \hat{y}_f$, as depicted in Fig. 4 with $K = 0$, the controller is no longer affected by the resonance behavior.

When the first three flexible modes of the system are compensated, the transfer from the feedback control effort, u_{fb} , to the compensated measurements, $\tilde{y}$, changes as seen in Fig. 6. The resonances and anti-resonances corresponding to the first three flexible modes are completely removed from

the transfer. This allows a redesign of the feedback controller for an increased control bandwidth.

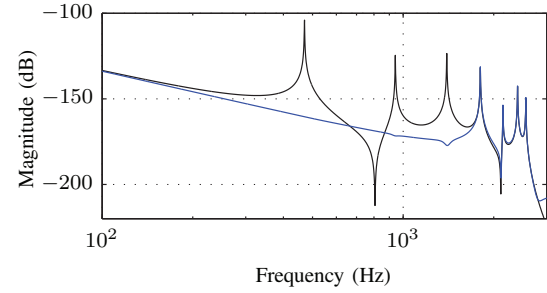


Fig. 6: Transfer function seen by the feedback controller with (blue) and without (black) measurement compensation.

Measurement compensation renders the flexible dynamics unobservable to the feedback controller but does not remove them from the system. Consequently, they can be excited and are still present in the true performance output $e_p = r - y$.

B. Modal damping

The second application uses the estimated state values to close a partial state feedback loop around the plant, as shown in Fig. 4 with $C_f = 0$. With an appropriate gain K this loop can increase the modal damping of the estimated flexible modes. This effectively reduces the resonance peaks in the transfer function seen by the controller as depicted in Fig. 7. This allows an increase in feedback control bandwidth.

The reduction of the resonance peaks is generally less than the reduction achieved by measurement compensation. Thus, the bandwidth increase that is attained by a controller redesign is generally lower than the increase when measurement compensation is applied. However, active modal damping reduces the flexible dynamics present in the system, whereas measurement compensation only renders them unobservable.

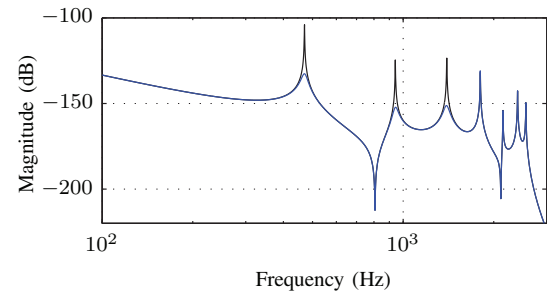


Fig. 7: Transfer function seen by the feedback controller with (blue) and without (black) active damping.

C. Combined result

The two applications of a state observer discussed previously, complement each other when applied to the system together. The downside of measurement compensation is the presence of the flexible dynamics on the true tracking error $e_p = r - y$. This is compensated by active damping which

actively reduces the flexible dynamics present in the system. Furthermore, the joint application allows at least the same increase in bandwidth as measurement compensation alone, which is more than that of active modal damping alone. In practice, the increase in bandwidth of the joint application is larger than that of measurement compensation alone. Due to modeling errors, measurement compensation cannot reduce the resonances completely, the additional damping in the joint application decreases the resonances even further and allows a larger bandwidth.

The complementary effect of both approaches will also be shown in Section VI where the largest disturbance rejection is obtained by the joint application of both methods.

VI. SIMULATION RESULTS

In this section the benefit of an observer on the flexible system described in Section III-A is shown. An observer is designed that estimates the rigid body mode and the first three flexible modes. A direct feedthrough compliance compensation is present for three more flexible modes. The last flexible mode is completely disregarded, to mimic the existence of unmodelled dynamics in the real system.

To show the increased disturbance rejection the reference $r(t)$ is set to zero and Gaussian, white, zero mean signals w and v with standard deviation, 10 mN and .15 nm respectively, are applied to the plant input and output. The Kalman observer (7) is designed using the same values. The spectral content of the tracking error is subsequently investigated. Four cases are considered:

- Original performance without observer
- Compensation of the measurement for the effects of the first three flexible modes
- Active modal damping for the first three flexible modes
- Combination of compensation (b) and damping (c)

Initially the same feedback controller is used in all cases, later a redesign for increased bandwidth is performed for cases b, c, and, d. The matrix K is chosen such that the loop gains for modal damping are 20 dB at the resonance frequency of each flexible mode. Fig. 8 displays the cumulative power spectral density (CPSD) of the compensated tracking error $e = r - \hat{y}$.

All applications reduce the flexible effects as the frequency content at the resonance frequencies is reduced. At frequencies below the resonance frequency the disturbance rejection has deteriorated. This low frequency increase will however be counteracted by the redesign of the controller later on.

The true tracking accuracy is not given by e but by e_p . Fig. 9 shows the CPSD of the real tracking error $e_p = r - y$. The plots for case a and c are unchanged as $\hat{y} = y$ in those applications. Cases b and d show a low frequency deterioration compared to Fig. 8. Case b also shows a large increase in spectral content at the resonance frequencies. This is expected as the modes are still present in the system. The additional damping in case d reduces this increase significantly.

The goal is improving the low frequency disturbance rejection by increasing the control bandwidth. New controllers

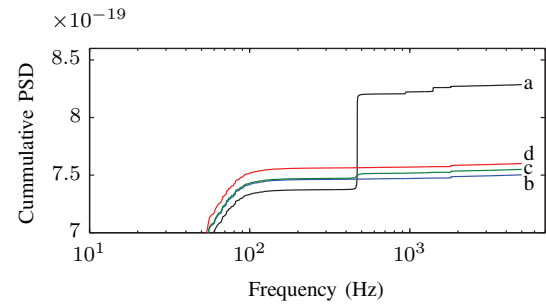


Fig. 8: Cumulative PSD of tracking error used in feedback ($e = r - \hat{y}$) for the four cases

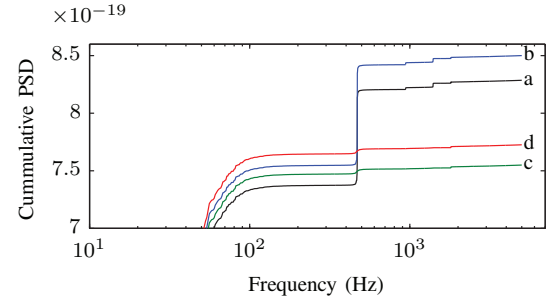


Fig. 9: Cumulative PSD of the true tracking error ($e_p = r - y$) for the four cases.

are designed with an increased bandwidth, for cases b, c, and d using the procedure of Section III-A. In cases b and d the dominant resonance is now the fourth flexible mode, thus the notch frequency is modified accordingly.

The resulting feedback control bandwidths are given in Table III. Note that the addition of active damping to measurement compensation does not yield an additional bandwidth increase. As this is a simulation example the compensation of the measurements is exact and the three resonances vanish completely. In practical applications measurement compensation will not completely remove the resonance peaks, due to model errors in the observer. Then, the added damping results in a further reduction that allows an even larger bandwidth.

Fig. 10 shows the CPSD of the true tracking error ($e_p = r - y$) when the new controllers are used. Clearly, the low frequency disturbance rejection has improved considerably for cases b, c, and d. Cases b and d show the largest improvement as the feedback control bandwidth is largest in those cases. The resonance frequencies are still present in the tracking error for case b. The combined application of compensation and damping in case d yields the best disturbance rejection.

VII. EXPERIMENTAL VALIDATION

A simple flexible motion system is used to validate the results obtained in the previous section. The experimental setup is shown in Fig. 11 and consists of two inertias

TABLE III: Feedback control bandwidth (0dB crossing of the open loop) for the four cases shown in Fig. 10

Original	Damping	Compensation	Damping and Compensation
39 Hz	90 Hz	130 Hz	130 Hz

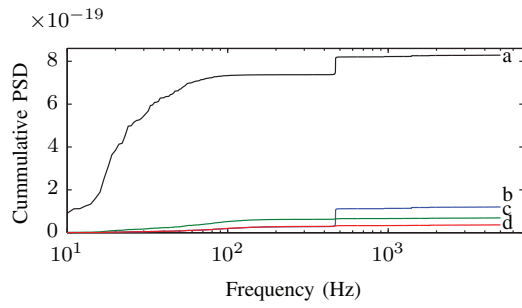
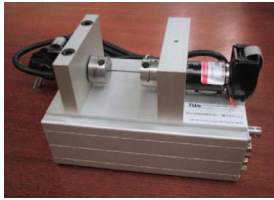


Fig. 10: Cumulative PSD of the true tracking error ($e_p = r - y$) for the four cases, with a redesigned feedback controller.

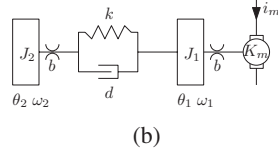
TABLE IV: Numerical values for the experimental setup.

Description	Symbol	Value
Inertia	J_1, J_2	$3.75\text{e-}6$ [kgm ²]
Friction	b	$1\text{e-}5$ [Nms/rad]
Spring	k	0.02656 [Nm/rad]
Damper	d	$3.125\text{e-}5$ [Nms/rad]
Motor constant	K_m	$4.4\text{e-}2$ [Nm/A]

connected by a small metal rod. The right inertia is connected to an electrical motor and the rotation of both inertias is measured by incremental encoders. The numerical values of the components are given in Table IV. Because the system only



(a)



(b)

Fig. 11: Experimental setup (a) and free body diagram (b)

contains one flexible mode there are no discarded dynamics and thus, $T_{DC} = 0$. A Kalman observer is designed and the noise covariance matrices are chosen as $Q = \text{diag}(.1, .1, 1, 1)$ and $R = 5 \cdot 10^{-5}$.

By measuring the transfer function experienced by the feedback controller the methods are validated. The experiments are done by injecting noise at the output of the feedback controller and performing closed loop system identification. A constant velocity reference of 15 revolutions per second is used during the experiments to remove the effect of static friction.

Fig. 12 shows the compensated plant transfer function for the four cases. The frequency content at 15 Hz is an artefact of an imbalance in the inertias. Modifying the reference velocity results in a accordingly changed frequency for the artefact. Measurement compensation removes the anti-resonance and almost completely compensates the resonance. Active damping only reduces the resonance and the combined application indeed provides the best result.

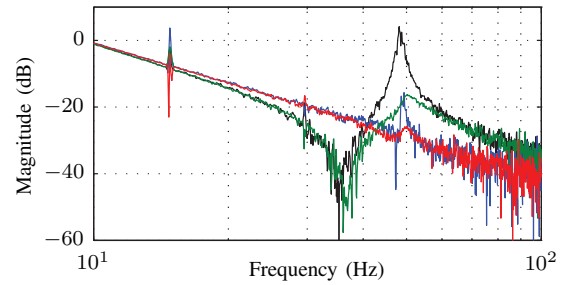


Fig. 12: Transfer function seen by the feedback controller for the four cases. a: black, b: blue, c: green, d: red

VIII. CONCLUSION

In this paper two applications of a state observer added to the existing control loop of a high precision flexible system have been presented. Both measurement compensation and active modal damping reduce the effects of flexible dynamics and facilitate a considerable increase of the feedback control bandwidth. As a consequence, the low frequency disturbance rejection increases which improves the positioning accuracy for high precision systems.

The best result is obtained by applying measurement compensation and active modal damping simultaneously. Measurement compensation allows the largest increase in control bandwidth and the added damping ensures that the flexible behavior is reduced on the performance output.

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